Four Majid Indoor-Localization Protocols:
A Fair, Executable Reconstruction and Method Benchmark

Integrated technical report

8 August 2026

Abstract

This report integrates four paper-specific tasks rather than pooling unlike experiments: MPUrge–MAP receiver localization, original MPUrge virtual- transmitter (VT) construction, SMART–LEA refinement, and the two-room CNN fingerprinting task. Every numerical result is labelled either a completed replacement-scene benchmark, a diagnostic smoke result, or published context. Fairness is the paper-specific number and location of physical acquisitions, not a delay-only rule: noisy power, AoA, and complex CIR are legal when obtained from those same acquisitions. Hidden ray/VT IDs, query truth, clean side channels, and extra spatial supervision are forbidden. The report gives a beginner-reproducible recipe for every method appearing in a table, complete audited leaderboards for all four reconstructed protocols, and a release/audit ledger.

Release state. All four replacement-scene full runs are complete and independently audited. MPUrge–MAP contains 40,000 primary rows; SMART contains 720,000 rows; original MPUrge VT construction contains 24 scene/noise blocks and 123,760 stored query predictions; and the corrected CNN run contains 115,200 rows across two rooms. Earlier smoke and interrupted-run artifacts remain explicitly diagnostic or superseded and do not supply a full leaderboard value.

Contents

1 What may and may not be claimed

The four source papers use simulator projects, scene files, and random seeds that are not available in this workspace. Consequently, a number produced here is an *executable replacement-scene reconstruction*; it is not an exact reproduction of the paper's number and not an external state-of-the-art claim. Published values appear only as context and are never pooled with reconstruction rows.

The paper/reconstruction boundary is operational:

1. A printed paper fact is stated as such and tied to its protocol.
2. A missing choice—geometry, AP coordinate, corruption extractor, threshold, seed, or ambiguity—is disclosed as a reconstruction choice.
3. A protected query may score a frozen method but may not select a model, threshold, sensor channel, fallback, or ablation.
4. A method may use every noisy physical observable at the same permitted acquisitions. It may not add labelled positions or read latent simulator identities, clean geometry, the true obstacle label, or query coordinates.

1.1 The four distinct protocols

Protocol	Output	Paper-side budget	Replacement implementation	Evidence state
MPUrge–MAP	One static query coordinate	Uniform 0.5 m RP map; 100 non-RP queries; independent range AWGN and obstruction tests	$4 \times 5 \times 3$ m image-source room; 80 cell-centred RPs; 100 locked off-grid queries; 16 conditions	Full, audited
Original MPUrge	Unordered VT coordinate set per transmitter	Nearby surveyed RPs; four physical transmitters and 16 VTs; several noise realizations; exact scene/layout unavailable	Six disclosed 20×20 m variable-visibility scenes; four noise levels; same sparse survey/query acquisition rule; corrected $P = 5$ and matching brackets	Full, audited
SMART–LEA	One static coordinate refined from an initial pose and VT/Tx anchors	Irregular floorplan; four physical plus 16 virtual transmitters; spacings $0.25, 0.5, 1.5, 2.5$ m	25×25 m rectangle; four Tx and four wall images/Tx; 1,000 locked queries; 3 seeds; clean/noisy	Full, audited
CNN	One static coordinate or RP posterior	Per room: 20 RPs, 60 AP positions, 7 configurations (8,400 fingerprints); 1,600 off-grid queries	Same counts and room sizes; image-source/noisy-array receiver; corrected grouped split, independently corrupted fit repeats, and strict validation leave-self logic	Full, audited

Table 1: The tasks have different outputs and supervision budgets; their errors are not one common leaderboard.

1.2 Shared acquisition and metric contract

Latent paths first synthesize a noisy array impulse response,

$$y_a[n] = \sum_{\ell} \alpha_\ell e^{-j(2\pi/\lambda)\mathbf{p}_a^\top \mathbf{u}_\ell} \mathbf{1}[n = n_\ell] + w_a[n], \qquad w_a[n] \sim \mathcal{CN}(0, \sigma^2). \tag{1}$$

Peak extraction yields a token $z_\ell = (\widehat{r}_\ell, \widehat{P}_\ell, \widehat{\mathbf{u}}_\ell)$ and leaves the noisy complex CIR available. Missing detection removes the whole token, so delay, power, AoA, dropout, and CIR noise are coupled. Survey and query acquisitions use independent draws. Latent source/ray identity and reflection order are never estimator features.

For a query truth $\mathbf{x}_b$ and prediction $\widehat{\mathbf{x}}_b$,

$$e_b = \|\widehat{\mathbf{x}}_b - \mathbf{x}_b\|_2, \quad \overline{e} = B^{-1}\sum_b e_b, \quad \mathrm{RMSE} = \sqrt{B^{-1}\sum_b e_b^2}. \tag{2}$$

Median and 90th percentile are also retained. Pairwise uncertainty uses the same physical blocks for both methods: with $\Delta_b = e_b^{\mathrm{baseline}} - e_b^{\mathrm{candidate}}$, a positive $\overline{\Delta}$ means the candidate is better. Blocks, not augmentation duplicates, are resampled.

2 Protocol I: corrected MPUrge–MAP

2.1 Frozen reconstruction

Every method receives the same 80 survey locations and the same 100 locked off-grid query locations. The 16 cells are ideal, nine range-AWGN levels $\sigma_r \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5\}$ m, five individual objects, and all objects. Each cell has one materialised survey tensor and an independently corrupted query tensor shared verbatim across methods. Learned augmentation corrupts the 80 stored fingerprints only (1,280 rows; zero new simulator coordinates). Selection uses 64 whole RP groups for fit and 16 for validation. Seven neural methods use seeds 211, 337, and 461. The 100 query labels are opened only for final scoring.

The observable channels are delay, noisy received power, noisy receive AoA, and noisy 8-element CIR. Zero-range-noise cells use the frozen high-SNR receiver setting (80 dB); range-AWGN cells use 25 dB for other modalities. That sensitivity choice is part of the replacement protocol and is a reason not to extrapolate its multimodal margin directly to hardware.

2.2 Common equations

For sorted query set $Q = \{q_a\}$ and RP set $R_i = \{r_{ib}\}$, corrected MPUrge uses

$$\delta_{ab} = \alpha|q_a - r_{ib}| + (1-\alpha)d_{\mathrm{pat}}(a,b), \tag{3}$$

$$d_{\mathrm{pat}}(a,b) = \binom{2p+1}{2}^{-1} \sum_{u<v} |, |W_a(u) - W_a(v)| - |W_{ib}(u) - W_{ib}(v)| \,| \,, \tag{4}$$

$$C_i = \frac{|\mathcal{M}_i|}{\max(|Q|, |R_i|, 1)}, \qquad s_i = \frac{1}{C_i|\mathcal{M}_i|}\sum_{(a,b)\in\mathcal{M}_i} \delta_{ab}. \tag{5}$$

Mutual-nearest matches are processed in increasing composite δ, and a new match may not reverse the delay order of an accepted match. The baseline uses $p = 6$, $\alpha = 0.7$, normalized pattern distance, and inverse-score interpolation over the three best finite RPs:

$$w_i = \frac{s_i^{-1}}{\sum_{j\in K} s_j^{-1}}, \qquad \widehat{\mathbf{x}} = \sum_{i\in K} w_i \mathbf{x}_i. \tag{6}$$

No finite evidence returns the survey-coordinate centroid for every method.

Three reusable decoders are

$$D_i^{\mathrm{Ch}} = \tfrac{1}{2}\left[|Q|^{-1}\sum_{q\in Q}\min_{r\in R_i}|q-r| + |R_i|^{-1}\sum_{r\in R_i}\min_{q\in Q}|r-q| \right], \tag{7}$$

$$w_i^{(T)} = \frac{\exp[-(s_i - s_{(1)})/T]}{\sum_{j\in K}\exp[-(s_j - s_{(1)})/T]}, \tag{8}$$

$$p^{(t+1)} = (1-\gamma)p^{(t)} + \gamma G^\top p^{(t)}, \qquad \widehat{\mathbf{x}} = \sum_i p_i^{(L)}\mathbf{x}_i, \tag{9}$$

where G is the normalized six-neighbour RP graph. The frozen EvoMDP row uses $E_i = \sum_{k=1}^{8} w_k f_k(Q, R_i)/a_k$ for eight delay-set distances: Chamfer, quantile Wasserstein, gap Wasserstein, log-cardinality ratio, partial assignment, local pattern, MCA complement, and moment difference.

For multimodal matching,

$$c_{ab} = |r_a - r_b| + \lambda_\theta \arccos(\mathbf{u}_a^\top \mathbf{u}_b) + \lambda_P |P_a - P_b|/10, \tag{10}$$
$$z_{ab} = c_{ab}/(1 + c_{ab}), \qquad S(z) = (1 - z)^2(1 + 2z). \tag{11}$$

The Beta-shaped row symmetrically averages nearest pair costs weighted by $S(z)$ and adds a normalized cardinality mismatch. "Beta-shaped" is the correct name: the implemented polynomial is not an exact Beta$(2, 5)$ survival integral. The coherent row uses the receiver's active-tap array projection distance D_i^{ADP} plus $\lambda_d D_i^{\mathrm{Ch}}$. Survival–CIR combines $D_i^{\mathrm{surv}} = aD_i^{\mathrm{Beta}} + bD_i^{\mathrm{ADP+ToA}} + cD_i^{\mathrm{Ch}}$; residual graph diffuses this score over the RP graph.

PointNet/Deep-Sets encode tokens using

$$h_\ell = \phi(z_\ell), \qquad e(Z) = [\mathrm{mean}_\ell\, h_\ell, \max_\ell h_\ell, \log(1 + |Z|)]. \tag{12}$$

CAEZ-style classification uses $p_i = \mathrm{softmax}(g)_i$ and $\widehat{\mathbf{x}} = \sum_i p_i \mathbf{x}_i$. A residual head uses $\widehat{\mathbf{x}} = \mathrm{clip}\{\mathbf{a} + \rho \tanh f_\theta(e, \mathbf{a}, d)\}$. RRLE predicts each frozen expert's error $\widehat{e}_k$ and mixes its four coordinates with $\pi_k \propto e^{-\widehat{e}_k/T}$. Genuine cross-attention uses

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(QK^\top/\sqrt{d} + M\right) V, \tag{13}$$

with padding mask M and no path-order positional encoding. The self-attention candidate model applies the same masked operation within each set, pools, then scores candidates equivariantly. Shuffle, padding, and candidate-permutation tests passed.

2.3 Recipe index for all 25 rows

ID	Artifact key	Beginner-reproducible recipe
M1	`majdi_mpurge_map_corrected_prose`	Compute corrected s_i above; inverse-weight the best three RPs.
M2	`majdi_mca_eps1.6_k1`	For each query delay, find its nearest RP delay; sum $(1.6 - d)^2$ only when $d < 1.6$ m; return the highest-scoring RP.
M3	`majdi_original_mpurge_P5_corrected`	Repeat M1 with total $P = 5$ (half-window 2), $\alpha = .5$, unnormalized pattern, no MAP coverage penalty.
M4	`ablation_A_chamfer_inverse_top3`	Replace only M1's score with exact symmetric Chamfer; retain inverse top-three and the common fallback.
M5	`ablation_B_temperature_softmax_only`	Retain M1 score; replace only inverse weights by development-selected temperature weights $w_i^{(T)}$.
M6	`ablation_C_path_count_empty_only`	Retain M1 and multiply finite scores by $1 + \eta\,||Q| - |R_i||/\max(|Q|, |R_i|)$; give a finite disclosed cost to empty references.
M7	`ablation_D_chamfer_temperature_pathcount`	Combine M4's Chamfer, M5's softmax, and M6's count/empty rule.
M8	`subset_consensus_chamfer`	Localize the full set and delete-one subsets (or all size-three subsets for at most six paths); keep hypotheses near their coordinate-wise median and average.
M9	`graph_diffusion_corrected`	Convert M1 scores to a temperature posterior, run L steps of six-neighbour diffusion G, and take its coordinate mean.
M10	`frozen_evomdp_rank`	Scale and mix the eight frozen delay distances f_k, temperature-decode the best frozen K RPs; do not refit the genome.
M11	`beta_survival_multimodal`	Form range/power/AoA pair costs c_{ab}, apply $S(z)$, symmetrize nearest supported costs, add path-count mismatch, inverse-decode three RPs.
M12	`coherent_adp_toa`	Add coherent active-tap CIR/array distance to weighted delay Chamfer; inverse-decode the best three.
M13	`survival_cir_toa`	Development-select the nonnegative mixture of M11, M12, and Chamfer, then inverse-decode.
M14	`assigned_vt_consensus_aoa`	Cluster survey endpoints $\mathbf{v} = \mathbf{x}_i + r\mathbf{u}$ without IDs; query proposals are $\mathbf{v} - r\mathbf{u}$; return the strongest multi-path DBSCAN consensus.
M15	`residual_graph_multimodal`	Apply M9's graph diffusion to M13's multimodal score instead of M1.
M16	`delay_feature_extratrees`	ExtraTrees on 17 delay quantiles, 7 gap quantiles, count, mean, standard deviation, minimum, and maximum; fit only stored calibration variants.
M17	`multimodal_feature_extratrees`	Add 7 power quantiles, weighted AoA mean/spread, and 7 normalized CIR summaries to M16; refit the tree ensemble.
M18	`rrle_moe`	Obtain M1, M4, M10, and M9 coordinates; predict their errors from delay/diagnostic/disagreement features; softmax-mix the experts.
M19	`pointnet_delay_direct`	Encode unordered scalar-delay tokens by shared MLP plus masked mean/max; directly regress normalized (x, y).
M20	`pointnet_multimodal_direct`	Use M19 with token $[r, P, u_x, u_y, u_z]$.
M21	`caez_grid_probability_delay`	Encode the delay set, output 80 RP logits, train by cross-entropy, and return the posterior coordinate mean.
M22	`residual_to_analytic_delay`	Feed delay-set encoding, M1 pose, and M1 diagnostics to a bounded residual head.

ID	Artifact key	Beginner-reproducible recipe
M23	`pointnet_analytic_reranker_delay`	Shortlist with M1; separately PointNet-encode query/reference delay sets plus coverage/score/RP coordinate; softmax-average candidates.
M24	`cross_attention_multimodal`	Let each query path attend to each candidate's reference paths using noisy range/power/AoA; masked-pool and score candidates.
M25	`candidate_self_attention_multimodal`	Self-attend within query/reference sets, pool invariantly, then apply a permutation-equivariant candidate attention/reranking head.

2.4 Complete pooled leaderboard

Rank	Method key	Mean	RMSE	Median	P90
1	`beta_survival_multimodal`	0.2528	0.3731	0.1627	0.5544
2	`survival_cir_toa`	0.2601	0.3789	0.1716	0.5649
3	`residual_graph_multimodal`	0.2745	0.4061	0.1940	0.5036
4	`candidate_self_attention_multimodal`	0.3349	0.4837	0.2156	0.6983
5	`cross_attention_multimodal`	0.3434	0.4868	0.2406	0.6836
6	`assigned_vt_consensus_aoa`	0.3463	0.8276	0.0238	1.3415
7	`pointnet_multimodal_direct`	0.4373	0.5317	0.3697	0.8041
8	`multimodal_feature_extratrees`	0.4480	0.5359	0.3866	0.8158
9	`coherent_adp_toa`	0.7746	0.9484	0.6739	1.5223
10	`rrle_moe`	1.0445	1.2987	0.8504	2.1244
11	`ablation_C_path_count_empty_only`	1.0589	1.3381	0.8213	2.2701
12	`majdi_mpurge_map_corrected_prose`	1.0638	1.3429	0.8075	2.2723
13	`ablation_B_temperature_softmax_only`	1.0721	1.3488	0.8269	2.2389
14	`subset_consensus_chamfer`	1.0752	1.3369	0.8409	2.2020
15	`pointnet_analytic_reranker_delay`	1.0961	1.2809	0.9760	2.0162
16	`ablation_A_chamfer_inverse_top3`	1.1077	1.3979	0.8655	2.3055
17	`graph_diffusion_corrected`	1.1084	1.3510	0.9460	2.1038
18	`residual_to_analytic_delay`	1.1252	1.3333	0.9803	2.0701
19	`ablation_D_chamfer_temperature_pathcount`	1.1345	1.4118	0.9387	2.2924
20	`frozen_evomdp_rank`	1.1353	1.4131	0.9120	2.2993
21	`delay_feature_extratrees`	1.2039	1.4055	1.0774	2.2369
22	`majdi_mca_eps1.6_k1`	1.3143	1.8017	0.8718	3.1994
23	`majdi_original_mpurge_P5_corrected`	1.3160	1.5544	1.2030	2.4810
24	`caez_grid_probability_delay`	1.3387	1.5261	1.1798	2.3893
25	`pointnet_delay_direct`	1.3509	1.5156	1.2350	2.2620

All values are metres over 1,600 identical condition/query blocks. The best three pooled methods are feature-rich; Beta-survival reduces mean error versus corrected Majid by 0.8110 m, query-block bootstrap 95% CI $[0.7224, 0.9033]$ m. Self-attention reduces it by 0.7289 m, independently audited CI $[0.6479, 0.8180]$ m; cross-attention reduces it by 0.7204 m, CI $[0.6415, 0.8052]$ m. Assigned-VT's 2.38 cm median but 1.34 m P90 reveals rare association failures.

Pure components do not establish a delay-only analytic win: A (Chamfer only) has mean 1.1077 m, reduction -0.0440 m and CI $[-0.1088, 0.0201]$ m; B (softmax only) 1.0721 m, reduction -0.0084 m and CI $[-0.0149, -0.0016]$ m; C (count/empty only) 1.0589 m, reduction 0.0049 m and CI $[-0.0036, 0.0133]$ m; D (all three) 1.1345 m, reduction -0.0708 m and CI $[-0.1373, -0.0044]$ m. Multimodal PointNet beats delay PointNet by 0.9136 m (CI $[0.7931, 1.0371]$ m), and multimodal ExtraTrees beats delay ExtraTrees by 0.7559 m (CI $[0.6433, 0.8742]$ m). The large gain is therefore mainly sensor-enabled; the attention rows show viability, not architecture-only causality.

3 Protocol II: original MPUrge VT construction

3.1 Corrected task and evidence boundary

This protocol estimates an *unordered set of VT coordinates*, not a receiver coordinate. Transmitter identity is observable because elementary transmitters use separable waveform channels; it is retained throughout matching, grouping, support counting, and evaluation. True VT coordinates, cardinality, ray identity, and query truth are never primary-method inputs. The paper's total window $P = 5$ is implemented as half-width $p = 2$. Current conflicts are resolved in increasing composite dissimilarity, and both the literal printed crossing test and geometric order-reversal test are recorded. Both overlapping-star and conflict-resolved connected-component grouping are bracketed. Beta support is swept until the retained set is empty.

The completed full design contains 24 independent scene/noise blocks: six scene seeds times $\sigma_r \in \{0, 0.25, 1, 3\}$ m, 36 survey positions, 20 held-out queries, 90 differentiable-assignment steps with two restarts, 120 epochs for each self-supervised set model, and a 31×31 downstream localization grid. All 24 blocks completed atomically. The independent audit verified 6,188 metric rows, 123,760 query predictions, the raw candidate/association evidence, 48 checkpoints, and all 86 manifest entries. The earlier interrupted 7/24-block attempt is retained only as non-result history.

The completed diagnostic smoke used one scene, $\sigma_r \in \{0, 0.25\}$ m, 20 survey positions, six queries, eight neural epochs, 12 assignment steps, and a 15×15 grid. Its one-scene intervals are necessarily degenerate.

3.2 Construction and scoring equations

For nearby survey anchors $\mathbf{x}_i$, match unordered indirect ranges with the corrected MPUrge dissimilarity from Protocol I (or the PBA L2 size- unification control). A conflict-compatible group $\mathcal{G}$ is converted to a VT by robust range trilateration,

$$\widehat{\mathbf{v}}_{\mathcal{G}} = arg\min_{\mathbf{v}\in\mathbb{R}^2} \sum_{(i,a)\in\mathcal{G}} \left(\|\mathbf{v}-\mathbf{x}_i\|_2 - r_{ia}\right)^2 . \tag{14}$$

For a candidate $\mathbf{v}$, support is the number of distinct anchors whose measured range agrees within the frozen gate. With mean support $\bar{s}$, the descriptive sweep retains $\{\mathbf{v} : s(\mathbf{v}) \geq \beta\bar{s}\}$ and increases β until this set is empty.

Multimodal inverse proposals are

$$\mathbf{v}_{ia} = \mathbf{x}_i + r_{ia}\mathbf{u}_{ia}. \tag{15}$$

Complete-link clusters with at least three anchors are power-weighted. The survival version keeps a cluster only when $\Pr[p > 0.15 \mid m, n] = 1 - F_{\mathrm{Beta}(1+m,1+n-m)}(0.15) \geq 0.90$. The Bernoulli/RFS tracker updates a matched track existence by $q \leftarrow q + 0.35(1-q)$ and a missed track by $q \leftarrow 0.94q$; it emits tracks with $q \geq 0.55$ and support from at least three anchors.

Unknown-cardinality differentiable assignment tests candidate counts, uses a soft symmetric set loss

$$L_K = \frac{1}{N}\sum_i \left[\frac{1}{|R_i|}\sum_{r\in R_i} \mathrm{smin}_{k,\tau} |r - d_{ik}| + \frac{1}{K}\sum_k \mathrm{smin}_{r\in R_i,\tau} |d_{ik} - r|\right] + \mathrm{BIC}(K), \tag{16}$$

where $d_{ik} = \|\mathbf{x}_i - \mathbf{v}_k\|_2$ and $\mathrm{smin}_\tau(a) = -\tau \log \sum_j e^{-a_j/\tau}$. DeepSets and self-attention instead predict a variable-existence set and minimize only range/AoA reconstruction on the allowed survey; no VT labels are used.

Evaluation first maximizes gated bipartite match cardinality and only then minimizes matched distance. If TP, FP, and FN are the resulting counts,

$$\text{precision} = \frac{TP}{TP+FP}, \quad \text{recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2TP}{2TP+FP+FN}. \tag{17}$$

Symmetric Chamfer, Hausdorff, and downstream grid-localization error prevent a high F1 from hiding poor coordinates or an unusable VT set.

3.3 Recipe index for every diagnostic row

ID	Artifact key	Reproduction recipe
V1	`bernoulli_rfs_multimodal`	Process survey anchors sequentially; form range/AoA proposals, gate to existing tracks, update coordinate/existence, decay missed tracks, retain $q \geq .55$ with three-anchor support.
V2	`beta_marginal_survival_multimodal`	Cluster range/AoA proposals as above; integrate the observed support under the Beta posterior and retain survival probability at least .90.
V3	`aoa_power_inverse_consensus`	Complete-link cluster $\mathbf{x}_i + r\mathbf{u}$ proposals and power-weight each cluster; require at least three distinct anchors.
V4	`corrected_pba_order_components`	L2 size unification, geometric crossing, connected-component groups, trilateration, dynamic beta.
V5	`corrected_mpurge_order_components`	L1 MPUrge size unification, geometric crossing, connected components, trilateration, dynamic beta.
V6	`cycle_consistent_delay`	Repeatedly sample three anchors/paths, trilaterate, require range support on other anchors, complete-link cluster accepted hypotheses.
V7	`corrected_pba_printed_components`	V4 with literal printed crossing instead of geometric order crossing.
V8	`corrected_mpurge_printed_components`	V5 with literal printed crossing.
V9	`corrected_mpurge_order_star`	V5 with overlapping one-hop star grouping instead of connected components.
V10	`corrected_pba_order_star`	V4 with overlapping star grouping.

ID	Artifact key	Reproduction recipe
V11	`corrected_pba_printed_star`	PBA, printed crossing, star grouping.
V12	`corrected_mpurge_printed_star`	MPUrge, printed crossing, star grouping.
V13	`diffassign_bic_multimodal`	Optimize L_K over candidate K, include noisy AoA reconstruction, choose K by survey-only BIC.
V14	`selfsup_deepsets_multimodal`	Shared token MLP, masked mean/max pooling, variable-existence VT decoder; fit survey range/AoA reconstruction only.
V15	`selfsup_attention_multimodal`	Replace V14's token interaction by masked self-attention without positional encoding; keep the same self-supervised loss.
V16	`diffassign_bic_delay`	V13 using surveyed delays and coordinates only.

3.4 Complete smoke leaderboard: diagnostic, not final

Rank	Method	F_1	Precision	Recall	Chamfer	Hausdorff	Loc. mean
1	`bernoulli_rfs_multimodal`	.9554	.9150	1.0000	.0559	.1986	.2864
2	`beta_marginal_survival_multimodal`	.8515	.7521	1.0000	.0807	.1882	.2853
3	`aoa_power_inverse_consensus`	.7729	.6390	1.0000	.0948	.2874	.3044
4	`corrected_pba_order_components`	.2572	.2910	.3125	8.7439	32.8345	5.1719
5	`corrected_mpurge_order_components`	.2556	.2897	.3125	8.6965	32.8345	5.0178
6	`cycle_consistent_delay`	.1584	.0905	.6875	3.2785	21.4026	.9159
7	`corrected_pba_printed_components`	.1442	.0874	.4375	4.3695	24.2796	.6892
8	`corrected_mpurge_printed_components`	.1412	.0851	.4375	4.2370	24.2796	.6799
9	`corrected_mpurge_order_star`	.0867	.0455	.9375	2.3526	17.8574	.3959
10	`corrected_pba_order_star`	.0860	.0451	.9375	2.3336	17.7446	.3724
11	`corrected_pba_printed_star`	.0758	.0395	.9375	2.6123	22.0860	.7048
12	`corrected_mpurge_printed_star`	.0756	.0394	.9375	2.6459	22.7884	.7124
13	`diffassign_bic_multimodal`	.0556	.0500	.0625	8.8376	21.4899	2.1195
14	`selfsup_deepsets_multimodal`	.0000	.0000	.0000	13.9164	22.7315	8.1569
15	`selfsup_attention_multimodal`	.0000	.0000	.0000	14.8455	30.7247	10.9501
16	`diffassign_bic_delay`	.0000	.0000	.0000	10.7496	21.8988	3.5806

Distances are metres. These numbers support triage only: the analytic multimodal RFS/survival/consensus branches deserve full comparison, while the eight-epoch neural failure is inconclusive and cannot justify rejecting set attention or DeepSets.

Audited full-run replacement

FULL_COMPLETE; independent artifact audit PASS. The disclosed replacement benchmark contains 24 scene/noise blocks (six independent scenes × four noise levels), 18 primary methods, 6,188 block-metric rows, and 123,760 stored query predictions. The true-cardinality oracle is diagnostic and excluded from every primary rank.

Table 6: Original-MPUrge VT construction, complete 24-block replacement-scene run. Brackets are deterministic scene-block bootstrap 95% CIs; distances are metres.

Rank	Method	Claim	P	R	F1 [95% CI]	Chamfer [95% CI]	Loc. [95% CI]
1	`two_sided_vt_registration`	primary	.5745	.5700	.5717 [.5470,.5971]	1.0275 [.8359,1.2351]	1.6455 [1.4025,1.8971]
2	`bernoulli_rfs_multimodal`	primary	.5034	.6328	.5562 [.5422,.5722]	.6528 [.5601,.7542]	1.4383 [1.2201,1.6695]
3	`beta_marginal_survival_multimodal`	primary	.4709	.6446	.5395 [.5193,.5559]	.7766 [.5604,1.0554]	1.4106 [1.3281,1.5024]
4	`diffassign_bic_multimodal`	primary	.4564	.5948	.5156 [.4681,.5527]	1.0577 [.8073,1.3598]	1.5711 [1.2798,1.8408]
5	`multipath_bundle_adjustment_noncheating`	primary	.5917	.4376	.5003 [.4394,.5690]	2.6358 [2.1029,3.2317]	2.0033 [1.7710,2.2145]
6	`aoa_power_inverse_consensus`	primary	.3218	.6556	.4217 [.4063,.4351]	.8241 [.7372,.9184]	1.6250 [1.3943,1.8515]
7	`diffassign_bic_delay`	primary	.3100	.4001	.3485 [.3206,.3768]	3.5305 [3.1710,3.9218]	1.7616 [1.6019,1.9025]
8	`corrected_mpurge_order_components`	primary	.1263	.1584	.1313 [.0985,.1626]	6.1565 [5.4041,6.7874]	3.2410 [2.5112,4.0659]
9	`corrected_pba_order_components`	primary	.1270	.1586	.1303 [.0900,.1679]	6.3960 [5.4433,7.3025]	3.3268 [2.4874,4.3483]
10	`corrected_pba_printed_components`	primary	.0993	.1540	.1158 [.0746,.1624]	5.9749 [5.2382,6.6700]	3.3874 [2.6305,4.1117]
11	`selfsup_deepsets_multimodal`	primary	.1294	.1035	.1140 [.0611,.1653]	3.6951 [2.7865,4.6029]	2.0763 [1.7915,2.2851]
12	`corrected_mpurge_printed_components`	primary	.0939	.1516	.1103 [.0719,.1517]	6.1957 [5.5566,6.8727]	3.4667 [2.7188,4.1718]
13	`selfsup_attention_multimodal`	primary	.0829	.0729	.0773 [.0584,.0963]	2.7805 [2.1840,3.3017]	1.9563 [1.7196,2.1943]
14	`corrected_mpurge_order_star`	primary	.0248	.6149	.0475 [.0427,.0521]	2.8050 [2.6703,2.9645]	1.8173 [1.6346,2.0295]
15	`corrected_pba_order_star`	primary	.0246	.6129	.0471 [.0412,.0523]	2.8160 [2.6963,2.9445]	1.8044 [1.6452,1.9972]
16	`cycle_consistent_delay`	primary	.0171	.5141	.0329 [.0301,.0351]	3.9508 [3.8342,4.0555]	2.3876 [2.1027,2.7186]
17	`corrected_mpurge_printed_star`	primary	.0165	.6274	.0321 [.0290,.0355]	3.1931 [3.0338,3.3776]	1.9752 [1.6450,2.2918]

Rank	Method	Claim	P	R	F1 [95% CI]	Chamfer [95% CI]	Loc. [95% CI]
18	`corrected_pba_printed_star`	primary	.0159	.6271	.0310 [.0290,.0336]	3.2148 [3.0433,3.3864]	1.9251 [1.6159,2.2206]
D	`diffassign_multimodal_oracle_cardinality_ablation`	oracle	.5750	.5750	.5750 [.5336,.6136]	1.0766 [.8558,1.3117]	1.5183 [1.2928,1.7289]

The printed/order and star/component ambiguities are separate native rows; no best-interpretation choice is hidden. Two-sided registration has the highest primary F1, Bernoulli RFS the smallest Chamfer distance, and beta-marginal survival the smallest localization error. Thus the run does not support one scalar "winner."

Table 7: Noise collapse for the leading transferable methods and the strongest native MPUrge row. Each cell is F1/localization mean (m), averaged over six independent scenes.

Method	$\sigma_r = 0$	.25	1	3 m
two-sided registration	.9769/.062	.8603/.150	.3337/1.272	.1157/5.098
Bernoulli RFS	.9603/.059	.8678/.167	.3456/.893	.0509/4.634
beta-marginal survival	.9253/.058	.7551/.159	.3812/.789	.0965/4.636
multimodal DiffAssign	.8068/.063	.7452/.283	.4482/.918	.0623/5.021
non-cheating multipath BA	.7469/.144	.7489/.214	.4786/1.803	.0269/5.852
AoA/power consensus	.8190/.061	.5487/.161	.2609/1.046	.0583/5.232
delay-only DiffAssign	.7383/.091	.5692/.193	.0865/1.500	.0000/5.263
MPUrge order/components	.3637/1.515	.1494/1.416	.0098/4.061	.0024/5.972

All families collapse at $\sigma_r = 3$ m, so the pooled means must not be read as uniform robustness. Multipath bundle adjustment selected the configured $K_{\max} = 7$ in 57 of 96 transmitter/block fits (other selected counts were $K = 3$: 2, $K = 4$: 3, $K = 5$: 10, $K = 6$: 24). Its result is therefore explicitly *cap-sensitive*; a larger, validation-frozen K sweep is needed before interpreting its poor Chamfer or high-noise localization as an algorithmic limit.

Provenance. Runner SHA-256 `4f1562bcfb3c72a4c92101367d70252ef30ecfacf4d4c33870b2e914596984ce`; immutable receiver SHA-256 `0ef4c590832a4d7c8bfac784d7b4c8e87e266bea1bb580ed805aa24d399999a4`; artifact-manifest SHA-256 `b0cb05c96fb06b5b9e12ff01e6f1b10e9e37d6b746aa2c77da2a1b9b654eb9b0`. The independent audit counted 234,215 candidate rows, 23,696 beta-set rows, 1,760,364 association rows, 123,760 query rows, 48 checkpoints, and 86 manifested artifacts. Exact evidence is under `research/four_paper_report/corrected_mpurge_vt_full_v3`. This remains replacement-scene evidence, not an exact run of the unavailable paper simulator.

4 Protocol III: SMART–LEA

4.1 Frozen full reconstruction

The paper describes an irregular two-dimensional floorplan with four physical transmitters and four first-order VTs per transmitter (20 anchors total). The replacement uses a disclosed 25×25 m rectangle, invented Tx coordinates, and four wall images per Tx. A single stored 100×100 survey at 0.25 m provides strict nested maps of 10,000, 2,500, 289, and 100 RPs for spacings $0.25, 0.5, 1.5, 2.5$ m. No coarse method sees dense-map rows. Each of seeds 29, 47, and 71 has 1,000 fixed off-grid queries in clean and matched-noisy conditions. Matched noise means 20 dB receiver SNR plus 0.25 m range, 2 dB power, and 5° direction perturbations after extraction. Learned selection holds out complete 5 m spatial blocks, then a fresh model is fitted at the permitted spacing.

SMART's literal primary input is an initial MCA coordinate, the coordinates of the 20 paper-known real/virtual transmitters, and the measured delay set. Richer map challengers instead receive only their declared same-acquisition features and RP coordinates. The separately labelled legacy four-solve row uses the observable physical-Tx channel partition; it is not Algorithm 1 and is excluded from primary winner claims.

4.2 Core equations and method recipes

At iteration t, SMART recalculates every anchor range and jointly matches all 20 values:

$$\widetilde{r}_j^{(t)} = \|\widetilde{\mathbf{x}}^{(t)} - \mathbf{v}_j\|_2, \qquad \widetilde{\mathbf{x}}^{(t+1)} = arg\min_{\mathbf{x}} \sum_{(a,j)\in\mathcal{M}^{(t)}} \left(\|\mathbf{x} - \mathbf{v}_j\|_2 - q_a\right)^2 . \tag{18}$$

The fixed-height correction evaluates $\sqrt{(x - v_{jx})^2 + (y - v_{jy})^2 + (z_r - v_{jz})^2}$ in both matching and least squares. Total $P = 5$ means half-window $p = 2$. The printed crossing test $(q_a - r_b)(q_c - r_d) < 0$ and geometric test $(q_a - q_c)(r_b - r_d) < 0$ are separate rows.

The additional analytic scores reuse MCA, Chamfer, graph diffusion, subset consensus, and EvoMDP from Protocol I. The combined delay score is

$$D_i = \beta \widetilde{D}_i^{\mathrm{Ch}} + (1-\beta)\left[\tfrac{1}{2}\widetilde{D}_i^{W_1} + \tfrac{1}{2}\left(1 - S_i^{\mathrm{MCA}} / \max_j S_j^{\mathrm{MCA}}\right)\right]. \tag{19}$$

For unit-normalized array-PDP $\mathbf{c}$, coherent distance is $D_i^{\mathrm{CIR}} = 1 - \mathbf{c}_q^\top \mathbf{c}_i$; delay–CIR uses $\eta \widetilde{D}_i^{\mathrm{Ch}} + (1-\eta)\widetilde{D}_i^{\mathrm{CIR}}$. Multimodal assignment uses Hungarian matching with

$$C_{ab} = |r_a - r_b|/(0.5\,\mathrm{m}) + |P_a - P_b|/(3\,\mathrm{dB}) + \arccos(\mathbf{u}_a^\top \mathbf{u}_b)/(10^\circ), \tag{20}$$

plus an unmatched-path penalty.

Survey-built VTs minimize a robust range residual over persistent tracks,

$$\widehat{\mathbf{v}}_k = \arg\min_{v_x, v_y, h \geq 0} \sum_{i \in \mathcal{T}_k} \rho\left(\sqrt{(x_i - v_x)^2 + (y_i - v_y)^2 + h^2} - r_{ik}\right). \tag{21}$$

The differentiable variant alternately Sinkhorn-normalizes $A_{iak} \propto e^{-|r_{ia} - d_{ik}|/\tau}$ and minimizes the robust assignment-weighted range loss. True VTs are used only after fitting for an audit error.

ID	Artifact key	Reproduction recipe
S1	`majid_mca_eps1`	MCA with $\epsilon = 1\,\mathrm{m}$, return the best RP; room centre on no evidence.
S2	`majid_smart_lea1_joint_totalP5_printed`	MCA start, one joint 20-anchor SMART iteration, total $P = 5$, printed crossing.
S3	`majid_smart_lea4_joint_totalP5_printed`	Repeat S2 for four iterations.
S4	`smart_lea4_joint_totalP5_geometric_cross`	S3 with geometric/order crossing only.
S5	`smart_lea4_joint_cited_halfwindow5_printed`	S3 with half-window 5 (total 11) to bracket the cited-parameter ambiguity.
S6	`smart_lea4_fixed_height_totalP5_geometric`	S4 with the known receiver height in three-dimensional anchor ranges.
S7	`legacy_per_tx_fusion_totalP5_printed`	Split by observable physical Tx, run four five-anchor solves, robustly fuse; diagnostic only.
S8	`corrected_mpurge_map_prefilter24`	Exact Chamfer shortlist 24 RPs, evaluate corrected coverage-penalized MPUrge only there, inverse top-three.
S9	`symmetric_chamfer_1nn`	Compute symmetric delay Chamfer and return its single best RP.
S10	`chamfer_temperature`	Development-select $K \in \{3, 5, 9\}$ and T, then softmax-average the top Chamfer RPs.
S11	`combined_delay_score`	Median-normalize Chamfer, Wasserstein, and MCA complement; select β on spatial development blocks.
S12	`subset_consensus`	Full/delete-one delay hypotheses, robust inlier filter, mean coordinate.
S13	`graph_diffusion`	Convert Chamfer scores to a grid posterior and diffuse through a normalized 3×3 kernel.
S14	`evomdp_rank_frozen_transfer`	Apply the frozen eight-distance genome without refitting its weights.
S15	`coherent_array_pdp_temperature`	Cosine distance between noisy normalized array-PDP vectors, temperature-decode the map.
S16	`range_power_aoa_assignment`	Hungarian range/power/AoA matching with unmatched penalty, temperature-average best RPs.
S17	`delay_cir_combined`	Development-select the normalized Chamfer/PDP-cosine mixture η.
S18	`pointnet_delay`	Shared delay encoder, masked mean/max/count pooling, direct coordinate head.
S19	`pointnet_range_power_aoa`	S18 with noisy $[r, P, \mathbf{u}]$ tokens.
S20	`query_reference_attention_delay`	Embed query and shortlisted reference delay sets; query-to-reference multihead attention; candidate softmax.
S21	`query_reference_attention_multimodal`	S20 with noisy range/power/AoA tokens.
S22	`candidate_pointnet_reranker_delay`	Independently PointNet-encode query/candidate sets and score encoding difference, RP coordinate, and Chamfer diagnostic.
S23	`candidate_path_cross_attention_multimodal`	Each query path cross-attends to each candidate's paths; masked-pool and softmax-average coordinates.
S24	`caez_style_delay_probability_mlp`	Fixed delay vector to one logit per allowed RP; posterior coordinate mean.
S25	`caez_style_cir_probability_mlp`	S24 with a 256-element pooled noisy array-PDP vector.
S26	`extra_trees_delay_features`	ExtraTrees on 20 padded delays, 19 gaps, count, mean, and standard deviation.
S27	`analytic_anchor_residual`	Chamfer-temperature anchor plus delay features/score margin/local spread; bounded 5 m tanh correction.
S28	`rrle_moe`	Route among Chamfer, ExtraTrees, delay PointNet, and analytic residual using spatially held-out predicted expert errors.
S29	`lea4_mpurge_track_vts`	Build survey-neighbour associations with corrected MPUrge, robustly fit persistent VTs, feed them to fixed-height LEA4.

ID	Artifact key	Reproduction recipe
S30	`lea4_diffassign_vts`	Initialize S29 VTs, refine with Sinkhorn range assignment on CUDA, feed the fitted set to the same LEA4.

4.3 Complete SMART macro leaderboard

The table macro-averages mean error across the four spacings, separately for clean and matched-noisy cells; the final column equally averages all eight condition/spacing cells. Every cell contains 3,000 protected query rows (three seeds times 1,000 queries). S7 is shown for completeness but marked diagnostic.

Rank	Method	Track	Clean	Noisy	All
1	`range_power_aoa_assignment`	multimodal	.3304	.5219	.4261
2	`delay_cir_combined`	multimodal	1.7888	1.7246	1.7567
3	`query_reference_attention_multimodal`	learned multimodal	1.4695	2.3953	1.9324
4	`coherent_array_pdp_temperature`	multimodal	2.1407	2.0927	2.1167
5	`symmetric_chamfer_1nn`	primary	2.4668	2.7267	2.5967
6	`corrected_mpurge_map_prefilter24`	primary	2.6957	3.0294	2.8626
7	`pointnet_range_power_aoa`	learned multimodal	2.7296	3.1246	2.9271
8†	`legacy_per_tx_fusion_totalP5_printed`	diagnostic	2.8405	3.4094	3.1249
9	`subset_consensus`	primary	2.9530	3.3626	3.1578
10	`evomdp_rank_frozen_transfer`	frozen transfer	3.0084	3.4497	3.2290
11	`combined_delay_score`	primary	2.8842	4.6757	3.7799
12	`chamfer_temperature`	primary	3.6735	4.1436	3.9086
13	`candidate_pointnet_reranker_delay`	learned	3.7571	4.3277	4.0424
14	`candidate_path_cross_attention_multimodal`	learned multimodal	4.0339	4.1794	4.1067
15	`query_reference_attention_delay`	learned	3.9112	4.3843	4.1477
16	`smart_lea4_fixed_height_totalP5_geometric`	analytic correction	3.8254	4.6683	4.2469
17	`smart_lea4_joint_totalP5_geometric_cross`	ambiguity bracket	3.8375	4.6767	4.2571
18	`analytic_anchor_residual`	learned hybrid	3.7423	5.0162	4.3792
19	`majid_smart_lea1_joint_totalP5_printed`	paper primary	3.9455	4.8167	4.3811
20	`majid_smart_lea4_joint_totalP5_printed`	paper primary	3.9393	4.8266	4.3829
21	`smart_lea4_joint_cited_halfwindow5_printed`	ambiguity bracket	4.0092	4.8396	4.4244
22	`majid_mca_eps1`	paper primary	4.0484	4.8740	4.4612
23	`graph_diffusion`	primary	4.3962	5.2339	4.8151
24	`caez_style_cir_probability_mlp`	learned multimodal	5.1453	5.1844	5.1648
25	`extra_trees_delay_features`	learned	3.6889	7.4714	5.5802
26	`lea4_mpurge_track_vts`	survey geometry	7.8734	7.8749	7.8742
27	`lea4_diffassign_vts`	survey geometry	7.6689	8.2406	7.9547
28	`rrle_moe`	learned hybrid	6.2895	9.9100	8.0998
29	`caez_style_delay_probability_mlp`	learned	7.5507	9.4283	8.4895
30	`pointnet_delay`	learned	9.2694	9.3757	9.3225

†Not eligible for the primary winner because it changes the paper's joint 20-anchor operation into four Tx-partitioned solves. Metres throughout.

The cell-wise winner changes with density and noise: fixed-height geometric SMART wins clean 0.25 m (0.0198 m) and 0.5 m (0.0562 m); range/power/AoA assignment wins clean 1.5 m (0.3197 m), clean 2.5 m (0.8576 m), noisy 1.5 m (0.4931 m), and noisy 2.5 m (1.0460 m); delay–CIR wins noisy 0.25 m (0.1338 m); multimodal query–reference attention wins noisy 0.5 m (0.2266 m). This is why a single macro rank must not replace per-density reporting.

4.4 All 30 methods in all eight SMART cells

Method	C-.25	C-.5	C-1.5	C-2.5	N-.25	N-.5	N-1.5	N-2.5
range_power_aoa_assignment	.060	.084	.320	.858	.241	.308	.493	1.046
delay_cir_combined	.088	.207	2.607	4.254	.134	.229	2.506	4.030
query_reference_attention_multimodal	.069	.129	.570	5.110	.181	.227	2.032	7.141
coherent_array_pdp_temperature	.164	.483	3.472	4.444	.165	.599	3.383	4.224
symmetric_chamfer_1nn	.102	.202	2.720	6.843	.320	.499	3.204	6.883
corrected_mpurge_map_prefilter24	.041	.111	3.820	6.810	.215	.404	4.443	7.056
pointnet_range_power_aoa	.661	1.109	6.830	2.318	1.373	1.844	5.409	3.872
legacy_per_tx_fusion_totalP5_printed	.071	.142	4.575	6.574	.776	1.413	4.852	6.597
subset_consensus	.044	.190	4.488	7.090	.282	.797	5.125	7.245
evomdp_rank_frozen_transfer	.048	.192	4.411	7.382	.318	.843	5.093	7.545
combined_delay_score	.145	.439	3.468	7.485	.346	1.883	7.416	9.058
chamfer_temperature	.060	.164	5.999	8.471	.303	.804	7.281	8.186
candidate_pointnet_reranker_delay	.059	.127	6.081	8.762	.337	1.045	7.018	8.911
candidate_path_cross_attention_multimodal	.064	.167	7.031	8.874	.213	.455	7.008	9.041
query_reference_attention_delay	.065	.191	6.682	8.707	.342	1.137	7.149	8.910
smart_lea4_fixed_height_totalP5_geometric	.020	.056	6.080	9.146	.845	1.787	6.687	9.355
smart_lea4_joint_totalP5_geometric_cross	.031	.068	6.089	9.163	.851	1.789	6.702	9.365
analytic_anchor_residual	.085	.223	6.117	8.544	1.510	2.824	7.455	8.275
majid_smart_lea1_joint_totalP5_printed	.048	.107	6.216	9.411	.911	1.862	6.838	9.656
majid_smart_lea4_joint_totalP5_printed	.037	.079	6.207	9.434	.922	1.843	6.812	9.729
smart_lea4_joint_cited_halfwindow5_printed	.037	.074	6.360	9.566	.889	1.852	6.930	9.688
majid_mca_eps1	.102	.238	6.385	9.469	.993	1.967	6.932	9.604
graph_diffusion	.060	.166	7.936	9.423	1.157	1.362	9.185	9.233
caez_style_cir_probability_mlp	.446	1.327	9.374	9.434	.501	1.510	9.336	9.391
extra_trees_delay_features	.618	1.853	5.042	7.243	5.625	6.840	8.412	9.009
lea4_mpurge_track_vts	4.095	4.910	9.004	13.484	4.723	5.780	9.763	11.233
lea4_diffassign_vts	3.904	4.104	9.839	12.829	3.929	5.739	10.414	12.879
rrle_moe	.265	1.456	11.000	12.438	5.666	8.531	12.875	12.568
caez_style_delay_probability_mlp	3.529	7.773	9.403	9.498	9.558	9.249	9.401	9.505
pointnet_delay	8.935	9.314	9.412	9.416	9.285	9.312	9.419	9.487

C denotes clean, N matched-noisy, and the suffix is map spacing in metres. Entries are mean position error in metres, read directly from the 240 summary rows in `smart_fair_full/results.json`.

Condition	Spacing	Winner	Mean	RMSE	Median	P90
clean	.25	fixed-height geometric LEA4	.0198	.0224	.0187	.0341
clean	.5	fixed-height geometric LEA4	.0562	.8452	.0187	.0342
clean	1.5	range/power/AoA assignment	.3197	.3715	.2813	.5385
clean	2.5	range/power/AoA assignment	.8576	1.2751	.6847	1.3894
noisy	.25	delay–CIR combined	.1338	.1524	.1240	.2276
noisy	.5	multimodal query–reference attention	.2266	.3131	.2038	.3850
noisy	1.5	range/power/AoA assignment	.4931	.5753	.4482	.8863
noisy	2.5	range/power/AoA assignment	1.0460	1.9119	.7351	1.5894

Table 11: Per-cell winner distributions. The clean 0.5 m LEA4 row has an excellent median/P90 but RMSE 0.8452 m, exposing a rare severe tail.

5 Protocol IV: two-room CNN fingerprinting

5.1 Printed protocol and corrected reconstruction

The paper uses Room A ($20 \times 8\,\mathrm{m}$) and Room B ($16 \times 12\,\mathrm{m}$), 20 RPs, 60 AP positions, and seven obstacle configurations. This grants $20 \times 60 \times 7 = 8{,}400$ labelled calibration acquisitions per room. It tests four obstacle-count conditions, two APs per condition, and 400 uniform points per condition: 1,600 locked off-grid queries per room. The printed input contains the nine ranges selected by *observed* power, all 20 RP coordinate triplets, and the AP coordinate, reshaped to 6×12.

The replacement preserves those counts and room sizes, but its disclosed AP positions, obstacle planes, and image-source propagation are not NIST Q-D. Each legal challenger receives no more than the printed 8,400 acquisitions; it may use noisy delay, power, receive AoA, and array CIR extracted there. It may not add positions, simulations, clean channels, ray/VT IDs, query truth, or hidden obstacle identity in the deployable bracket.

Within each AP/configuration group, 16 of 20 RP acquisitions fit and four validate. Validation pseudo-queries compare only with fit-map rows and explicitly exclude their source acquisition:

$$j \notin \mathcal{C}(Q_j), \qquad \mathcal{C}(Q_j) \subseteq \mathcal{I}_{\mathrm{fit}}. \tag{22}$$

Candidate-network training instead uses independent measurement-corrupted, nonempty repeats of fit acquisitions. A repeat may retain its original stored fit acquisition as a candidate so singleton AP/RP cells remain defined; this adds neither a spatial site nor a propagation simulation. Empty validation and locked queries are retained and use frozen finite fallbacks. Locked queries may use the complete acquired map only after all choices freeze.

Reference methods have two non-equivalent information brackets. The paper-matched sensitivity bracket supplies the true obstacle condition $R_i^{\mathrm{oracle}} = R_{aic_q}$ and is explicitly privileged. The deployable environment-blind bracket selects, per RP,

$$c_i^* = \arg\min_{c \in \{0,\dots,6\}} D(Q, R_{aic}), \qquad R_i^{\mathrm{blind}} = R_{aic_i^*}, \tag{23}$$

without exposing the label. Direct models use neither bracket and are emitted once as protocol-independent rows.

5.2 Equations and the 25-method recipe index

Fit-only min–max scaling converts observed-power-ranked delays to $\widetilde{d}_{1:9}$. The paper vector

$$v = [\widetilde{d}_{1:9}, \widetilde{\mathbf{x}}^{\mathrm{RP}}_{1:20}, \widetilde{\mathbf{a}}] \in \mathbb{R}^{72} \tag{24}$$

is reshaped column-major to $1 \times 6 \times 12$. The classifier outputs $p_i = \mathrm{softmax}(g)_i$ and $\widehat{\mathbf{x}} = \sum_i p_i \mathbf{x}_i$ with cross-entropy; the regressor minimizes the printed batch coordinate RMSE. Three 2×1 convolution blocks yield 36,148 classifier and 36,190 regressor parameters; the paper's reported regressor count differs by 21 and remains unresolved.

Map methods reuse MCA, corrected MPUrge, Chamfer, subset consensus, graph diffusion, and EvoMDP from Protocol I. Multimodal pair costs are

$$C_{ab}^{rP} = |r_a - r_b| + |P_a - P_b|/8, qquad C_{ab}^{r\theta} = |r_a - r_b| + 1.5 \arccos(\mathbf{u}_a^\top \mathbf{u}_b). \tag{25}$$

For active CIR taps, coherent ADP uses

$$\rho_n = \frac{|\mathbf{q}_n^H \mathbf{r}_n|^2}{\|\mathbf{q}_n\|_2^2 \|\mathbf{r}_n\|_2^2}, \qquad D_{\mathrm{ADP}} = 1 - \frac{\sum_n \sqrt{E_{qn} E_{rn}} \rho_n}{\sum_n \sqrt{E_{qn} E_{rn}}}. \tag{26}$$

Survey-built inverse consensus clusters $\mathbf{v} = \mathbf{x}_i + r\mathbf{u}$ and proposes query coordinates $\widehat{\mathbf{x}}_{\ell k} = \mathbf{v}_k - r_\ell \mathbf{u}_\ell$. Beta-marginal visibility contributes $\log p(m \mid n) = \log B(2 + m, 1 + n - m) - \log B(2, 1)$.

ID	Planned artifact key	Reproduction recipe
C1	`majid_cnn_classifier`	Build the printed 72-vector, column-major reshape, three conv blocks, 20-way cross-entropy, posterior RP mean.
C2	`majid_cnn_regressor`	Same input/backbone as C1, two-coordinate head, printed batch-RMSE loss.
C3–4	`majid_mca_eps1`, `majid_mca_eps2`	MCA at $\epsilon = 1$ and 2 m under the declared map bracket.
C5	`corrected_mpurge_map`	Coverage-corrected MPUrge, $p = 6$, normalized pattern, printed crossing, inverse top-three.
C6	`symmetric_chamfer_inverse3`	Symmetric delay Chamfer, inverse-weight the best three.
C7	`symmetric_chamfer_softmax3`	C6 with fixed $T = .03$ softmax over the best three.
C8	`subset_consensus`	Full and path-subset Chamfer hypotheses, median-radius inlier fusion.
C9	`graph_diffusion`	Rank-transform MPUrge evidence, diffuse over the two-nearest-RP graph, blend 75% analytic and 25% graph mode.
C10	`evomdp_frozen_transfer`	Frozen eight-distance mixture; fit only feature scales, then top-three temperature decode.
C11	`range_power_chamfer`	Symmetric nearest-neighbour matching with C^{rP}; top-three $T = .15$ head.
C12	`range_aoa_chamfer`	Same with $C^{r\theta}$.
C13	`dichasus_coherent_adp_8nn`	Noise-gated coherent ADP, spatially distinct best eight RPs, temperature barycentre.
C14	`assigned_vt_inverse_consensus`	Survey-build VTs from range/AoA proposals, form query inverse proposals, geometric-median consensus blended with a coarse delay pose.
C15	`beta_marginal_vt_survival`	One-to-one inverse-proposal/VT assignment plus Beta-integrated support and unmatched penalties, top-three decode.
C16	`pointnet_delay`	Shared scalar-delay encoder, masked mean/max/count, AP context, direct coordinate head.
C17	`pointnet_multimodal`	C16 with noisy range/power/AoA tokens.
C18	`set_attention_multimodal`	Two masked four-head self-attention blocks without positional encoding, invariant pooling, direct coordinate head.
C19	`candidate_pointnet_reranker`	Independently set-encode query/reference, add coordinate/AP/analytic score, guarded candidate softmax.
C20	`candidate_set_attention`	C19 with self-attended query/reference encoders.
C21	`genuine_candidate_cross_attention`	Each query path attends to candidate paths; pool $[Q, A, Q - A]$ with map/AP diagnostics and guarded logits.
C22	`caez_cir_probability_mlp`	Pool noisy CIR magnitude to 64 bins, concatenate AP coordinate, output 20 RP probabilities and their coordinate mean.
C23	`analytic_anchor_residual`	Environment-blind MPUrge anchor plus multimodal PointNet and score margin; bounded 0.25 tanh update.
C24	`survival_cir_toa_extratrees`	ExtraTrees on 32 CIR energy-survival samples, first ToA, path count, and AP coordinate.
C25	`rrle_map_aware_moe`	Freeze environment-blind MPUrge, Chamfer, and ADP experts; learn validation-error/disagreement router; softmax-mix coordinates.

Audited corrected full two-room CNN result

This is a replacement-scene reconstruction, not a numerical reproduction of the paper's unavailable NIST Q–D realization. It retains the printed two-room topology: 20 reference positions, 60 AP positions, seven obstruction conditions, 8,400 stored calibration fingerprints per room, and 1,600 locked off-grid queries per room. Every modality is extracted from the same noisy array-CIR acquisition; no extra calibration position, dense forward simulation, latent ray/VT identity, or query truth is available.

How to read the brackets. The paper-condition bracket supplies a reference method with the true matching obstruction-condition map and is therefore privileged but closest to the paper prose. The environment-blind bracket pools all seven stored condition maps and chooses through observed fingerprint similarity only. Protocol-independent rows either use no reference map or a VT map built from the allowed pooled survey. They are emitted once rather than copied under multiple labels.

Uncertainty. For Room A, error reductions are paired against the reconstructed Majid CNN classifier; for Room B they are paired against the reconstructed Majid CNN regressor. The 95% intervals resample the 1,600 physical off-grid queries (2,000 bootstrap replicates), preserving the pairing across methods and protocol brackets. A positive reduction means lower localization error than that room's baseline.

5.3 Room A

5.3.1 Paper-condition (privileged, paper-matched) map

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	Range–AoA Chamfer	1.3852	1.8524	1.1829	2.3282	4.3054 [4.1533, 4.4588]
2	Range–power Chamfer	2.3651	3.2779	1.7072	4.6695	3.3255 [3.1626, 3.5013]
3	Chamfer + softmax top-3	2.5916	3.6009	1.8017	5.6114	3.0990 [2.9244, 3.2757]
4	Subset consensus	2.6181	3.3297	2.1271	4.7784	3.0726 [2.9023, 3.2489]
5	Chamfer + inverse top-3	2.6255	3.3381	2.1280	4.8756	3.0651 [2.9090, 3.2297]
6	EvoMDP frozen transfer	2.6597	3.5898	2.0020	5.4929	3.0309 [2.8618, 3.2032]
7	Graph diffusion	2.7623	3.5877	2.1203	5.3828	2.9283 [2.7543, 3.1016]
8	Corrected MPUrge–MAP	2.7898	3.5492	2.2123	5.3936	2.9008 [2.7270, 3.0772]
9	Majid MCA ($\epsilon = 2$)	3.2452	4.5363	2.1481	6.9821	2.4454 [2.2282, 2.6462]
10	Majid MCA ($\epsilon = 1$)	3.6738	4.9617	2.6439	7.5535	2.0168 [1.7949, 2.2299]
11	DICHASUS coherent ADP 8-NN	6.5595	7.7142	5.9039	12.4914	-0.8689 [-1.0877, -0.6477]

5.3.2 Environment-blind deployable map

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	Range–AoA Chamfer	1.3594	1.8129	1.1503	2.3060	4.3312 [4.1761, 4.4940]
2	Candidate self-attention	1.8043	2.5008	1.3632	3.3836	3.8863 [3.7229, 4.0561]
3	Candidate PointNet reranker	1.8914	2.5001	1.4614	3.6621	3.7992 [3.6508, 3.9651]
4	Analytic-anchor residual	2.0529	2.6948	1.5706	3.9741	3.6377 [3.4814, 3.7985]
5	Range–power Chamfer	2.3483	3.2500	1.7140	4.6590	3.3423 [3.1682, 3.5048]
6	Genuine candidate cross-attention	2.5561	3.2803	1.9856	5.0736	3.1346 [2.9708, 3.2923]
7	Chamfer + softmax top-3	2.5973	3.6644	1.8328	5.4148	3.0933 [2.9098, 3.2672]
8	Subset consensus	2.6136	3.3630	2.1370	4.8214	3.0771 [2.9182, 3.2378]
9	Chamfer + inverse top-3	2.6231	3.3866	2.1423	4.8297	3.0675 [2.8994, 3.2313]
10	EvoMDP frozen transfer	2.6432	3.5558	2.0022	5.3550	3.0474 [2.8772, 3.2226]
11	Graph diffusion	2.8090	3.6392	2.2082	5.4235	2.8817 [2.7153, 3.0539]
12	Corrected MPUrge–MAP	2.8400	3.5932	2.3243	5.3661	2.8506 [2.6905, 3.0110]
13	RRLE map-aware MoE	3.1650	3.9311	2.6308	5.8754	2.5256 [2.3661, 2.6957]
14	Majid MCA ($\epsilon = 2$)	3.2935	4.5452	2.3638	6.9151	2.3971 [2.1986, 2.6163]
15	Majid MCA ($\epsilon = 1$)	3.6490	4.8863	2.7076	7.4079	2.0416 [1.8282, 2.2667]
16	DICHASUS coherent ADP 8-NN	6.3172	7.5292	5.6345	12.4300	-0.6266 [-0.8421, -0.4161]

5.3.3 Protocol-independent direct/pooled-vt

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	PointNet (multimodal)	0.7946	1.1940	0.5265	1.6120	4.8960 [4.7458, 5.0563]
2	Set attention (multimodal)	0.9826	1.6401	0.6070	1.9805	4.7080 [4.5606, 4.8546]
3	Beta-marginal VT survival	1.2856	1.6879	1.2067	1.8541	4.4050 [4.2498, 4.5609]
4	Survival-CIR + ToA ExtraTrees	2.2887	2.9545	1.8006	4.2849	3.4019 [3.2431, 3.5605]
5	CAEZ CIR probability MLP	2.3391	3.3159	1.6776	4.7325	3.3515 [3.1735, 3.5241]
6	Assigned-VT inverse consensus	2.3452	3.4846	2.1885	5.4577	3.3454 [3.1764, 3.5113]
7	PointNet (delay)	2.6855	3.3043	2.2548	4.6821	3.0052 [2.8455, 3.1675]
8	Majid CNN regressor	2.9745	3.6128	2.6509	5.0555	2.7161 [2.5622, 2.8789]
9	Majid CNN classifier	5.6906	6.5059	5.3885	9.9638	baseline

5.4 Room B

5.4.1 Paper-condition (privileged, paper-matched) map

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	Range–AoA Chamfer	1.2785	1.5671	1.1464	2.1074	1.6409 [1.5410, 1.7367]
2	Range–power Chamfer	2.1524	2.9409	1.5882	4.5505	0.7670 [0.6501, 0.8773]
3	Chamfer + softmax top-3	2.3930	3.3185	1.6412	5.0650	0.5264 [0.4074, 0.6510]
4	EvoMDP frozen transfer	2.5579	3.4257	1.8110	5.7050	0.3615 [0.2419, 0.4828]
5	Subset consensus	2.5794	3.1647	2.1630	4.9689	0.3401 [0.2419, 0.4362]
6	Chamfer + inverse top-3	2.5836	3.1768	2.1563	4.9991	0.3358 [0.2409, 0.4359]
7	Majid MCA ($\epsilon = 2$)	2.8507	3.9576	1.7117	7.0692	0.0687 [-0.0650, 0.2144]
8	Graph diffusion	2.9979	3.7706	2.3143	6.3797	-0.0784 [-0.1927, 0.0380]
9	Corrected MPUrge–MAP	3.0910	3.8058	2.4675	6.3008	-0.1716 [-0.2849, -0.0611]
10	Majid MCA ($\epsilon = 1$)	3.3114	4.5588	1.9887	8.2421	-0.3920 [-0.5393, -0.2348]
11	DICHASUS coherent ADP 8-NN	6.1726	7.0999	5.8874	11.2169	-3.2532 [-3.4326, -3.0697]

5.4.2 Environment-blind deployable map

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	Range–AoA Chamfer	1.1878	1.4717	1.0598	1.9352	1.7316 [1.6386, 1.8277]
2	Candidate self-attention	1.5963	1.9536	1.3488	3.0224	1.3231 [1.2292, 1.4212]
3	Candidate PointNet reranker	1.7242	2.1625	1.4132	3.2781	1.1953 [1.0988, 1.2925]
4	Analytic-anchor residual	1.9795	2.4520	1.6436	3.8925	0.9399 [0.8435, 1.0372]
5	Range–power Chamfer	2.0841	2.7278	1.5865	4.3740	0.8354 [0.7233, 0.9422]
6	Chamfer + softmax top-3	2.3097	3.1023	1.6103	5.1430	0.6097 [0.4829, 0.7286]
7	Genuine candidate cross-attention	2.3592	2.8988	2.0072	4.5078	0.5602 [0.4591, 0.6557]
8	EvoMDP frozen transfer	2.3686	3.1240	1.6900	5.3506	0.5508 [0.4439, 0.6670]
9	Subset consensus	2.5379	3.0517	2.2149	4.9569	0.3816 [0.2904, 0.4701]
10	Chamfer + inverse top-3	2.5450	3.0640	2.2056	4.9784	0.3744 [0.2817, 0.4671]
11	Graph diffusion	2.8798	3.5509	2.3349	5.8968	0.0396 [-0.0726, 0.1518]
12	Majid MCA ($\epsilon = 2$)	2.9266	4.0171	1.8149	6.8058	-0.0072 [-0.1489, 0.1283]
13	Corrected MPUrge–MAP	2.9844	3.6090	2.5193	5.7564	-0.0650 [-0.1747, 0.0406]
14	Majid MCA ($\epsilon = 1$)	3.2333	4.3432	2.1206	7.3808	-0.3139 [-0.4620, -0.1580]
15	RRLE map-aware MoE	3.5563	4.2168	3.1056	6.7095	-0.6369 [-0.7610, -0.5131]
16	DICHASUS coherent ADP 8-NN	5.5255	6.4280	5.1625	10.1594	-2.6061 [-2.7765, -2.4237]

5.4.3 Protocol-independent direct/pooled-vt

Rank	Method	Mean	RMSE	Median	P90	Reduction vs. room baseline [95% CI]
1	PointNet (multimodal)	0.8251	1.1871	0.5468	1.8212	2.0944 [2.0025, 2.1874]
2	Set attention (multimodal)	1.1233	1.8621	0.5904	2.7423	1.7961 [1.6957, 1.8910]
3	Beta-marginal VT survival	1.1657	1.3198	1.1824	1.6710	1.7538 [1.6620, 1.8492]
4	CAEZ CIR probability MLP	1.7625	2.3294	1.3386	3.5647	1.1569 [1.0427, 1.2581]
5	Survival-CIR + ToA ExtraTrees	2.0052	2.4271	1.7007	3.7775	0.9142 [0.8247, 1.0022]
6	Assigned-VT inverse consensus	2.5536	3.5186	2.4995	5.9157	0.3658 [0.2471, 0.4861]
7	PointNet (delay)	2.5853	3.0491	2.2567	4.7699	0.3341 [0.2448, 0.4224]
8	Majid CNN regressor	2.9194	3.4820	2.4935	5.4474	baseline
9	Majid CNN classifier	5.3789	5.9457	5.2338	8.6921	-2.4595 [-2.6040, -2.3266]

5.5 Audit and interpretation

The serialized evidence contains exactly 115,200 finite raw prediction rows, 25 unique numerical methods, and 72 room–protocol–method summaries. The independent audit recomputed every error and aggregate, verified the 11/16/9 protocol method topology, verified 1,600 queries for every room–protocol–method cell, and found no complete prediction vector copied across protocol labels. The frozen runner SHA–256 is `a5aa5f18b9807116df3bf54e0546a12b5dcc98a886c9cb447769f65bc45d9d90`; the frozen receiver SHA–256 is `10962bf2ab1edf3495879c2693f3e75ad1aeabf68c5af359d9f108645d3b5e67`. The tables compare algorithms within the reconstructed receiver. Published paper numbers must remain context columns rather than being pooled with these rows: matching the paper's counts does not recreate its unavailable geometry, ray realization, or exact NIST Q–D outputs.

The superseded `cnn_fair_smoke` predates strict leave-self validation and provenance corrections; none of its values enters this report. Published context only: Room A classifier 1.82 m, regressor 1.84 m, MCA-1 2.58 m; Room B classifier 2.14 m, regressor 1.47 m, MCA-1 3.65 m. These are paper values, not replacement rows.

6 Compact causal and multilevel evidence

This section is intentionally limited to at most two pages and is not inserted into any static-protocol leaderboard. Temporal filtering has a different information budget; active multilevel calibration has a different output question.

6.1 Confirmed positive causal sequential result

G2 wraps the frozen corrected-MPUrge RP score vector. Rank scores form the emission and surveyed RP distances form the transition:

$$\ell_t(i) = \exp\left[-\beta\,\frac{\mathrm{rank}(s_t(i))}{N-1}\right], \tag{27}$$

$$T_{ij} \propto \exp[-\|\mathbf{x}_i - \mathbf{x}_j\|_2^2/(2\sigma_T^2)], \tag{28}$$

$$p_t = \mathrm{normalize}\{\ell_t \odot (T^\top p_{t-1})\}, \qquad \widehat{\mathbf{x}}_t = \sum_{i\in \mathrm{top}\,K(p_t)} \frac{p_t(i)}{\sum_{j\in K} p_t(j)}\mathbf{x}_i. \tag{29}$$

Earlier frames froze $\beta = 8$, $\sigma_T = .6\,\mathrm{m}$, zero extra stay weight, and $K = 5$ before three shifted 200-frame confirmation segments were opened. The predictor uses only past belief, current analytic scores, and RP coordinates; no future observation or query coordinate enters it.

Method	Mean	Median	P90
G2 causal analytic HMM/particle bank	.3802	.2793	.7907
All-path PointNet	.8121	.3405	1.3887
Corrected MPUrge single-frame	.9654	.3503	2.4194

Across 1,350 matched rows / 270 physical frames, G2 reduces mean error versus PointNet by 0.4319 m, frame-block CI $[0.3541, 0.5137]$ m, and versus static MPUrge by 0.5852 m, CI $[0.4958, 0.6783]$ m. This is a confirmed positive result, but only for ordered trajectories.

6.2 Honest causal multilevel calibration: useful, not a confirmed win

Nine nested acquisition policies choose 12, 20, or 30 of 72 survey candidates without reading any unselected fingerprint: spatial farthest, Chebyshev, polynomial Leja, Smolyak, Sobol, GP variance, Bayesian quadrature, FEM variation, and surrogate disagreement. Each policy renders the missing map with its development-frozen polynomial/FEM/RBF choice and localizes 36 protected queries. The causal acquisition oracle recorded zero unselected reads, no dense-map selection, no query access, and nested paths.

Budget	Lowest observed policy	Mean	Spatial mean	Reduction	95% CI
12	spatial farthest	1.5251	1.5251	.0000	$[0, 0]$
20	causal FEM variation	1.4744	1.5462	.0718	$[-.1534, .3035]$
30	causal GP variance	1.3434	1.4271	.0837	$[-.1002, .2674]$

The positive point estimates at budgets 20 and 30 have intervals crossing zero, so the correct conclusion is neutral rather than a win. An earlier "goal-surplus" result is retracted as calibration-savings evidence because it read every unselected fingerprint. Recent quadrature/multifidelity ideas are therefore not promoted in the headline report: under the honest oracle they did not show a great confirmed result.

7 Release and audit ledger

Evidence block	Status	Auditable scale	Release ruling
MPUrge–MAP full	PASS with disclosed post-run source mutation	40,000 primary rows; 33,600 learned seed rows; 16 conditions; 25 methods	Publish as replacement-scene evidence. All 34 focused checks and invariance/edge tests pass.
SMART full	FULL_COMPLETE; integrity PASS	720,000 raw rows; 24 scenarios; 30 methods	Publish as replacement-scene evidence; retain legacy Tx-partition row as diagnostic.
Original MPUrge smoke	Quick smoke complete	1 scene; 2 noise levels; 16 primary rows; all 12 stored hashes recomputed	Retain only as diagnostic history; the audited full run supersedes its ranks.
Original MPUrge full	FULL_COMPLETE; independent audit PASS	24/24 blocks; 18 primary methods; 123,760 query rows; 86 manifest entries	Publish as replacement-scene evidence; keep the true-cardinality row oracle-only and disclose high-noise collapse and BA cap sensitivity.
CNN legacy smoke	SUPERSEDED	Old quick artifact	Do not quote; it predates leave-self and provenance fixes.
CNN corrected full	FULL_COMPLETE; 21-check independent audit PASS	115,200 rows; 25 unique methods; 72 summaries; 1,600 queries per room/bracket row	Publish with privileged, environment-blind, and protocol-independent rows separated; never compare replacement-scene values numerically with published NIST Q-D values.
G2 causal	PASS	1,350 rows / 270 frames	Publish separately as sequential evidence, never in the static table.
Causal multilevel	Integrity PASS; no confirmed gain	27 policy-budget summaries; 36 protected queries each	Publish the honest neutral finding; never restore the invalid oracle win.

7.1 Hash and provenance resolution

The MAP manifest preserves receiver SHA-256 `7e9909312d9f9330e5e3d37f7ab3962732b2500503e2c1113e09c3f1d18da8f6`; the SMART manifest preserves its imported receiver SHA-256 beginning `5a4867f3d1363c54`. A concurrent task changed the shared receiver after those completed runs and restored its semantics without byte-identical text; the current file hashes to `10962bf2ab1edf34 95879c2693f3e75ad1aeabf68c5af359d9f108645d3b5e67`. The manifests must not be rewritten to imply that current bytes generated old rows. Completed result/checkpoint hashes remain intact; the mismatch is resolved by disclosure, not by altering run outputs.

7.2 Exact evidence paths

All paths below are relative to the project root.

Purpose	Path
MAP aggregates	`research/four_paper_report/corrected_mpurge_map_full/aggregate_metrics.json`
MAP paired intervals	`research/four_paper_report/corrected_mpurge_map_full/paired_bootstrap_vs_corrected_majid.json`
MAP raw/seed rows	`research/four_paper_report/corrected_mpurge_map_full/raw_predictions.jsonl`; `learned_seed_predictions.jsonl`
MAP audit/provenance	`research/four_paper_report/corrected_mpurge_map_full/focused_result_audit.json`; `provenance_resolution.json`
SMART result/raw/audit	`research/four_paper_report/smart_fair_full/results.json`; `raw_predictions.jsonl.gz`; `integrity_audit.json`
VT diagnostic smoke	`research/four_paper_report/corrected_mpurge_vt_smoke/results.json`; `SHA256SUMS.txt`
VT audited full	`research/four_paper_report/corrected_mpurge_vt_full_v3/results.json`; `all_method_leaderboard.csv`; `block_audit.json`
CNN full result/audits	`research/four_paper_report/cnn_fair_corrected_full_final2/results.json`; `integrity_audit.json`; `independent_integrity_audit.json`
CNN corrected runner	`experiments/four_paper_report/run_cnn_fair.py`; `test_cnn_fair.py`; `audit_cnn_full_results.py`
Four-paper freeze/audits	`research/four_paper_report/protocol_freeze.md`; `mpurge_protocol_audit.md`; `smart_cnn_protocol_audit.md`
Method dependency inventory	`research/four_paper_report/requested_method_inventory.md`
G2 causal evidence	`research/campaign_8plus4/g2_replication_results.json`; `g2_replication_rows.jsonl`

Multilevel evidence	`research/multilevel_idp/confirmation_results.json`; `confirmation_rows.jsonl`

7.3 Minimal reproducibility checklist

1. Materialize and hash survey/query observations before training. Verify independent survey/query corruption and absence of forbidden identity fields.
2. Enforce the exact paper-specific acquisition count. At sparse SMART spacing, subset the stored dense acquisition before any fit; for CNN, never exceed the printed 8,400 acquisitions per room.
3. Fit normalization and choose all hyperparameters on grouped calibration only. Exclude the pseudo-query's source row for validation. Independently corrupted nonempty training repeats may retain their stored fit acquisition as a candidate when needed to keep singleton AP/RP cells defined; disclose this policy and add no spatial site or propagation call.
4. Freeze configuration, source hashes, seed list, fallbacks, brackets, and method registry before opening locked queries. Keep learned seeds separate in raw evidence before ensembling.
5. Save prediction coordinates or VT sets, scoring truth, physical block keys, runtime, feature provenance, map-information bracket, checkpoints, and hashes. Reconstruct every reported error mechanically from raw rows.
6. Run empty/no-evidence, padding, path-shuffle, candidate-shuffle, and finite-output tests. Cluster paired intervals by physical query/AP/scene, not augmentation row.
7. Label every number as paper context, full reconstruction, diagnostic smoke, or oracle sensitivity. Never use a completed task's number to fill another task's missing table.

8 Bottom line

On the completed MAP replacement, the best dependable row is the same-acquisition range/power/AoA Beta-shaped survival adapter at 0.2528 m mean, versus 1.0638 m for corrected prose-consistent Majid. Survival–CIR and the multimodal residual graph are close; permutation-invariant self-attention and genuine cross-attention are also strong. Controlled feature ablations show that most of this margin comes from richer noisy sensing, not attention alone.

On SMART, the dominant method depends sharply on map density and receiver noise. Fixed-height geometric LEA4 is exceptionally accurate on dense clean maps but has a rare severe tail at 0.5 m; multimodal assignment dominates sparser maps, delay–CIR wins the densest noisy map, and multimodal attention wins noisy 0.5 m. The all-cell table must accompany any headline.

On original MPUrge VT construction, two-sided registration has the highest primary VT-set F1 (0.5717), Bernoulli RFS has the smallest Chamfer distance (0.6528 m), and beta-marginal survival has the smallest localization error (1.4106 m); all degrade severely at the strongest noise, so no single robust winner is claimed. On the corrected CNN reconstruction, multimodal PointNet wins both rooms at 0.7946 and 0.8251 m mean error. Its advantage is paired and large within the replacement scene, but it is not a numerical claim against the paper's unavailable NIST Q-D scene. Separately, the causal analytic G2 filter is a confirmed positive sequential extension, whereas honest active multilevel calibration is neutral under paired uncertainty.